

Supplementary Figures: Strain–Doping Coupling in Graphullerene

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(Dated: June 19, 2026)

Companion to the main manuscript on coupled (ϵ, δ) response in qHP C_{60} . Figures adopt the observables vocabulary of Capobianco *et al.* [1] and its Supplementary Information (Electron SI): band-edge π -DOS (SI Fig. S4, purple valence / teal conduction), inverse participation ratio IPR (SI Eq. S8), electronic coupling J (SI Eq. S11), Marcus λ (vertical–adiabatic split), and Franck–Condon weighted density FCWD (MolFC generating-function approach, SI Sec. S11). Our tetramer Exp. 4–9 audits use the same labels at PBE+D3; absolute J and λ are *not* directly comparable to the Koopmans-compliant periodic supercells of Ref. 1, but provide internally consistent (ϵ, δ) trends. **DATA:** Fig. S5 λ panels update when ver-

tical single-points converge; until then adiabatic IP/EA only.

TABLE I. Exp. 5 tetramer energies and strain sensitivities (PBE+D3).

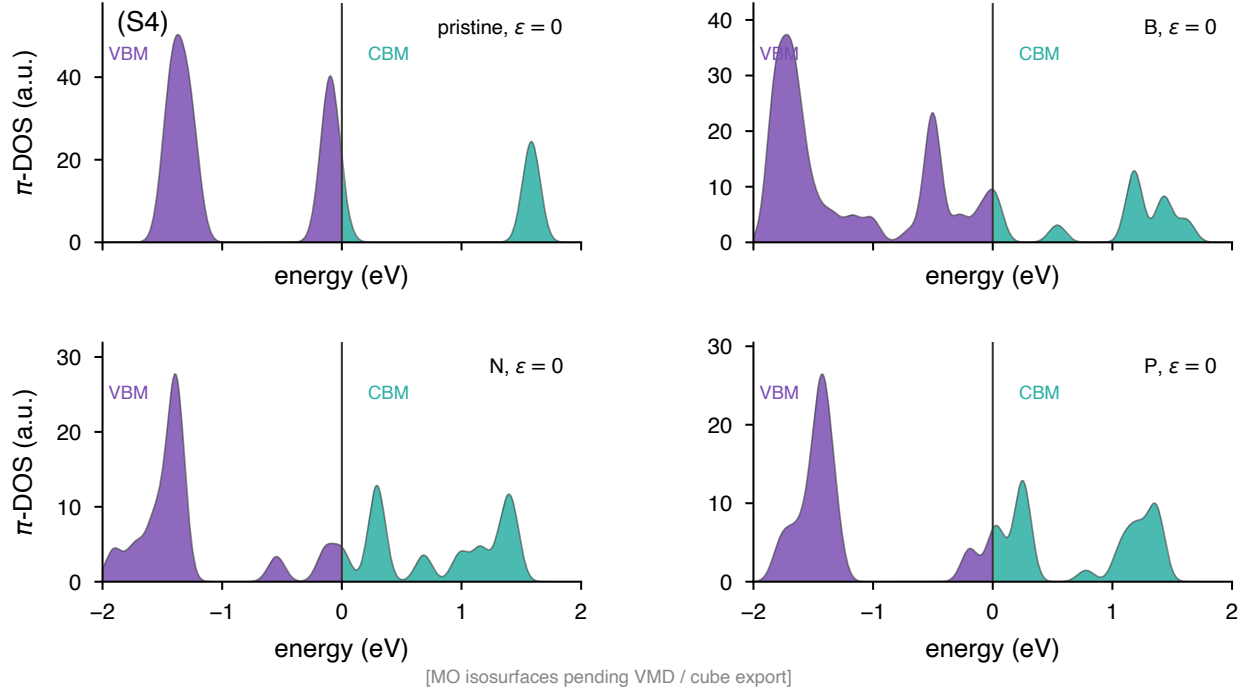
System	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_f (eV/dopant)
Pristine	−1363.551	−1363.551	−1.0	—
B-doped	−1328.490	−1328.409	+56.9	+159.0
N-doped	−1414.381	−1414.405	−303.2	−230.5
P-doped	−1370.496	−1370.489	−19.1	−31.5

[1] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa,

Nano Letters **24**, 7891 (2024), key reference from refer-info.md - polaron transport mechanisms.

TABLE II. DFT campaign ledger (converged CP2K tasks archived with inputs; main-text use).

Campaign	Model	N_{conv}	Main text	Notes
Structure probe (Exp. 1)	$2 \times \text{C}_{60}$ strain	5	—	superseded by tetramer grids
Doping scan (Exp. 2)	single C_{60}	11	—	superseded
Electronics (Exp. 3)	$2 \times \text{C}_{60}$	24	—	superseded by Exp. 7
IPR/ J (Exp. 4)	tetramer	8 (4 in 2×2)	Fig. S6, main inset	pristine/B $\epsilon=0, +3\%$ $\times$
Strain energies (Exp. 5)	tetramer	24	Table I, α	rigid strain
Optimal grid (Exp. 6)	tetramer	20	—	SI archive
PDOS/gaps (Exp. 7)	tetramer	22	Fig. S4; main gap/DOS panels	
Geometry (Exp. 8)	tetramer relax	11	SI inputs	reference geometries
Charged (Exp. 9)	tetramer $q = \pm 1$	23	SI Fig. S5	λ pending
Size scaling (Exp. 10)	$n \times \text{C}_{60}$	40	main $\mathcal{S}(n)$ panel	280 Ry for $n \geq 6$

FIG. 1. π -projected DOS at $\epsilon=0$ for pristine and B/N/P tetramers (Exp. 7, PBE+D3). Band-edge π -DOS follows Electron SI Fig. S4 colour scheme (VBM purple, CBM teal); MO isosurfaces [pending: VMD + cube export].

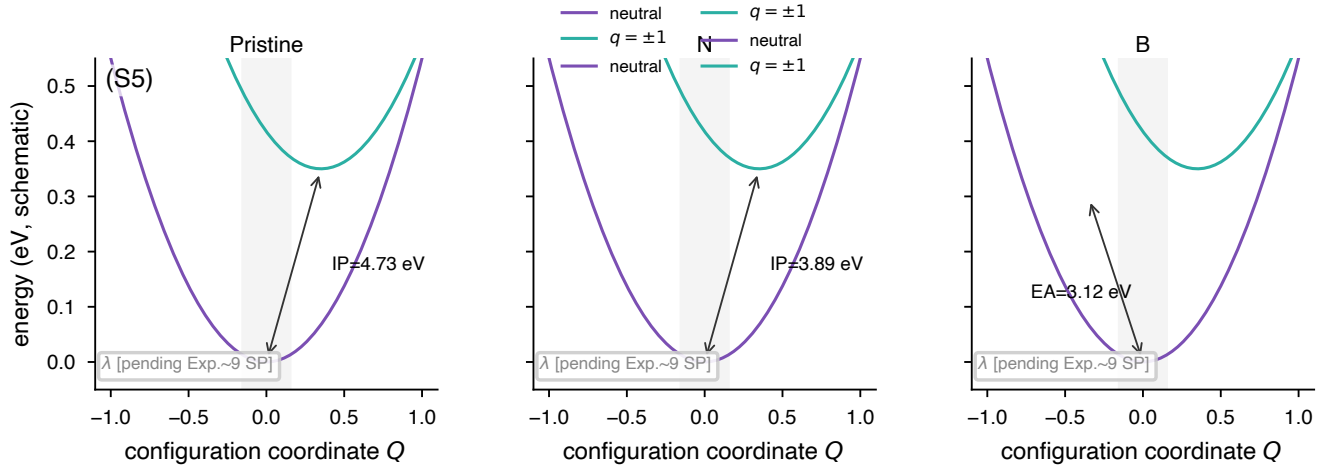


FIG. 2. Marcus configuration-coordinate schematic (Capobianco-style). Configuration-coordinate schematics following the Marcus vertical-adiabatic decomposition of Ref. 1; verified adiabatic IP/EA from converged Exp. 9 GEO.OPT pairs (pristine IP, N IP, B EA). $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ remains [pending: vertical SP at neutral geometry].

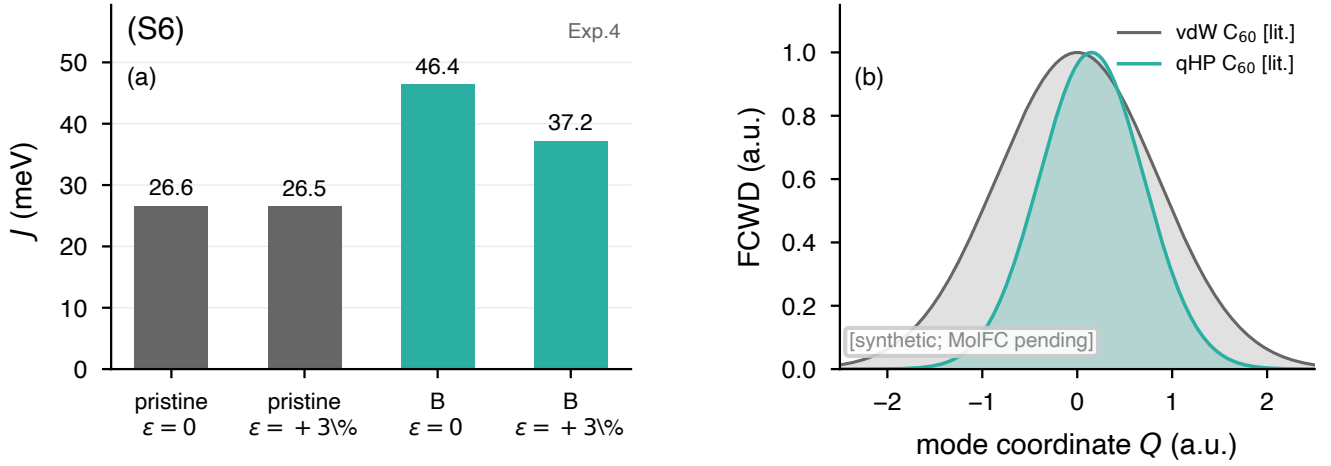


FIG. 3. Transport-adjacent observables: (a) nearest-neighbor J on the Exp. 4 2×2 factorial (pristine/B $\times$ $\epsilon=0, +3\%$, PBE+D3); (b) illustrative FCWD envelope (Electron SI MolFC/GF methodology [synthetic placeholder; tetramer MolFC pending]).